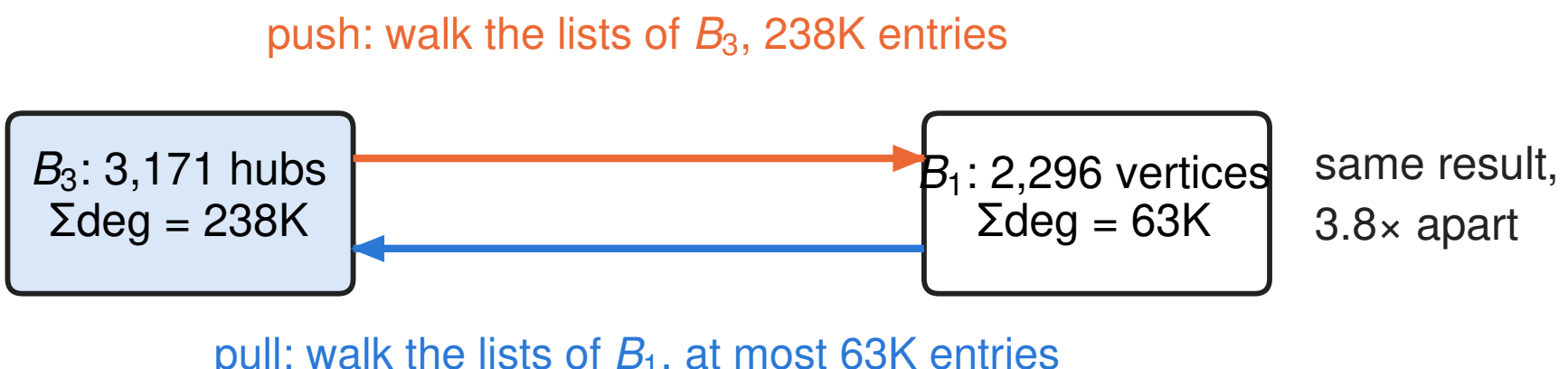
(a) relational: one filter per plan edge, in the pass direction

cost: scan S to build, probe each row of T;
the join scans S and T anyway

(b) graph: either direction over the adjacency index



same result,
3.8× apart

the direction is a choice per transfer, and the cheaper one is known exactly